

LAB Assignment No 8

Smart Temperature & Humidity Monitoring System with ESP32 and ThingSpeak

In this lab, students will design and implement a basic IoT-based environmental monitoring system using an ESP32 microcontroller and a DHT22 sensor. The system will measure temperature and humidity in real time and apply simple decision logic to control an LED when predefined threshold values are exceeded. Additionally, sensor data will be transmitted to the cloud using ThingSpeak, where it can be visualized through live graphs.

This lab introduces students to key IoT concepts including sensor interfacing, wireless communication, cloud integration, and real-time data monitoring, forming a foundation for more advanced IoT applications.

Lab Objectives

- To interface DHT22 sensor with ESP32
- To monitor temperature and humidity in real time
- To control an LED based on threshold conditions
- To send sensor data to ThingSpeak cloud
- To visualize IoT data using cloud graphs

Learning Outcomes

After completing this lab, students will be able to:

- Understand sensor interfacing with ESP32
- Implement conditional logic in embedded systems
- Use WiFi for IoT communication
- Send data to cloud using ThingSpeak API
- Analyze real-time data using cloud visualization
- Build a complete end-to-end IoT system

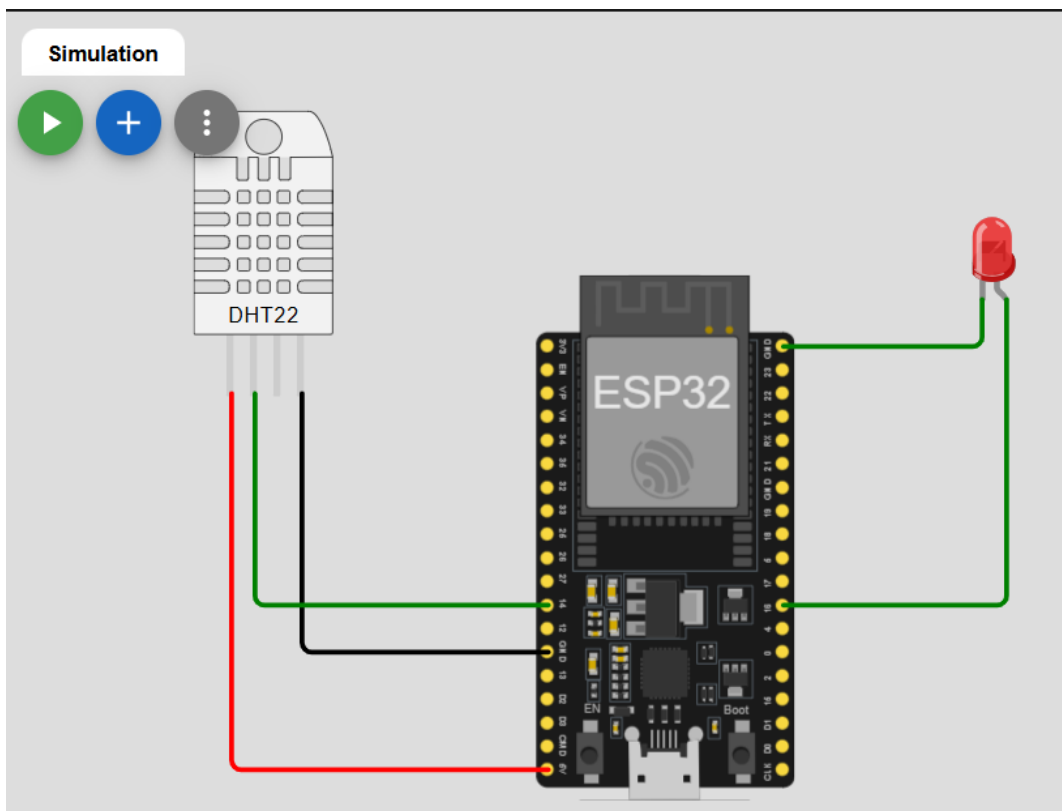
Required Hardware

- ESP32 Development Board
- DHT22 Temperature & Humidity Sensor
- LED
- 220Ω Resistor
- Breadboard & Jumper Wires

Required Software / Simulation

- Arduino IDE
- Wokwi Simulator
- ThingSpeak Account

Circuit Diagram



ThingSpeak Setup

1. Create an account on ThingSpeak
2. Create a new **Channel**
3. Add fields:
 - Field 1 → Temperature
 - Field 2 → Humidity
4. Copy:
 - Channel ID
 - Write API Key

Program Code

```
#include <WiFi.h>
#include <ThingSpeak.h>
#include "DHT.h"

// Channel details
unsigned long channelID = 3365699;
const char* writeAPIKey = "4O1QPV6ZU8S8KB8C";

// WiFi client
WiFiClient client;

// DHT setup
#define DHTPIN 14
#define DHTTYPE DHT22
#define LED_PIN 16

DHT dht(DHTPIN, DHTTYPE);

// WiFi credentials (Wokwi default)
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// Threshold values
float tempThreshold = 30.0;
float humThreshold = 60.0;

void setup() {
  Serial.begin(115200);
```

```
pinMode(LED_PIN, OUTPUT);
digitalWrite(LED_PIN, LOW);

dht.begin();

// Connect WiFi
WiFi.begin(ssid, password);
Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nConnected!");

// Initialize ThingSpeak
ThingSpeak.begin(client);
}

void loop() {
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    if (isnan(temperature) || isnan(humidity)) {
        Serial.println("Failed to read from DHT sensor!");
        return;
    }

    Serial.print("Temp: ");
    Serial.print(temperature);
    Serial.print(" °C | Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");

    // LED Control Logic
    if (temperature > tempThreshold || humidity > humThreshold) {
        digitalWrite(LED_PIN, HIGH);
        Serial.println("⚠️ ALERT: LED ON");
    } else {
        digitalWrite(LED_PIN, LOW);
        Serial.println("✅ Normal: LED OFF");
    }

    // Send data to ThingSpeak
    ThingSpeak.setField(1, temperature);
```

```
ThingSpeak.setField(2, humidity);

int response = ThingSpeak.writeFields(channelID, writeAPIKey);

if (response == 200) {
  Serial.println("Data successfully sent to ThingSpeak");
} else {
  Serial.print("Error sending data. HTTP code: ");
  Serial.println(response);
}

// ThingSpeak requires 15 seconds delay
delay(15000);
}
```

Working Principle

1. ESP32 reads data from DHT22
2. Compares values with threshold:
 - Temp > 30°C
 - Humidity > 60%
3. If condition TRUE → LED ON
4. Data is sent to ThingSpeak every 15 seconds
5. Graph is updated in real-time

Wokwi Simulation



Expected Simulation Output



Students should observe:

- Serial monitor showing values
- LED turning ON/OFF
- Sensor values changing

ThingSpeak Output

Expected Cloud Output

 Students should observe:

- Temperature graph
- Humidity graph
- Real-time updates

LAB Tasks:

Students are required to:

Task 1

Implement ESP32 + DHT22 interfacing

Task 2

Apply LED control logic based on threshold

Task 3

Send data to ThingSpeak cloud

Task 4

Verify cloud graphs

Task 5 (Extension)

Change threshold values and observe behavior

Deliverables

Students must submit:

- Source code
- Screenshot of Wokwi simulation
- Screenshot of ThingSpeak graphs
- Short explanation (1–2 pages)

Viva Questions

1. What is DHT22 sensor?
2. Difference between DHT11 and DHT22
3. What is IoT cloud?
4. Why delay is required in ThingSpeak?
5. What is API key?
6. Difference between HTTP and MQTT
7. How ESP32 connects to WiFi?

```
1  #include <WiFi.h>
2  #include <ThingSpeak.h>
3  #include "DHT.h"
4
5  // ✅ Updated Channel details
6  unsigned long channelID = 3301731;
7  const char* writeAPIKey = "A3VKBHCTA5US6SVK";
8
9  // WiFi client
10 WiFiClient client;
11
12 // DHT setup
13 #define DHTPIN 14
14 #define DHTTYPE DHT22
15 #define LED_PIN 16
16
17 DHT dht(DHTPIN, DHTTYPE);
18
19 // WiFi credentials (Wokwi default)
20 const char* ssid = "Wokwi-GUEST";
21 const char* password = "";
22
23 // Threshold values
24 float tempThreshold = 30.0;
25 float humThreshold = 60.0;
```

```

27 void setup() {
28     Serial.begin(115200);
29
30     pinMode(LED_PIN, OUTPUT);
31     digitalWrite(LED_PIN, LOW);
32
33     dht.begin();
34
35     // Connect WiFi
36     WiFi.begin(ssid, password);
37     Serial.print("Connecting to WiFi");
38
39     while (WiFi.status() != WL_CONNECTED) {
40         delay(500);
41         Serial.print(".");
42     }
43
44     Serial.println("\n✅ Connected to WiFi!");
45
46     // Initialize ThingSpeak
47     ThingSpeak.begin(client);
48 }
49
50 void loop() {
51     float temperature = dht.readTemperature();

```

```

54     // Check sensor
55     if (isnan(temperature) || isnan(humidity)) {
56         Serial.println("❌ Failed to read from DHT sensor!");
57         return;
58     }
59
60     // Print values
61     Serial.print("⌚ Temp: ");
62     Serial.print(temperature);
63     Serial.print(" °C | 💧 Humidity: ");
64     Serial.print(humidity);
65     Serial.println(" %");
66
67     // LED Control Logic
68     if (temperature > tempThreshold || humidity > humThreshold) {
69         digitalWrite(LED_PIN, HIGH);
70         Serial.println("⚠️ ALERT: LED ON");
71     } else {
72         digitalWrite(LED_PIN, LOW);
73         Serial.println("✅ Normal: LED OFF");
74     }
75
76     // Send data to ThingSpeak
77     ThingSpeak.setField(1, temperature);

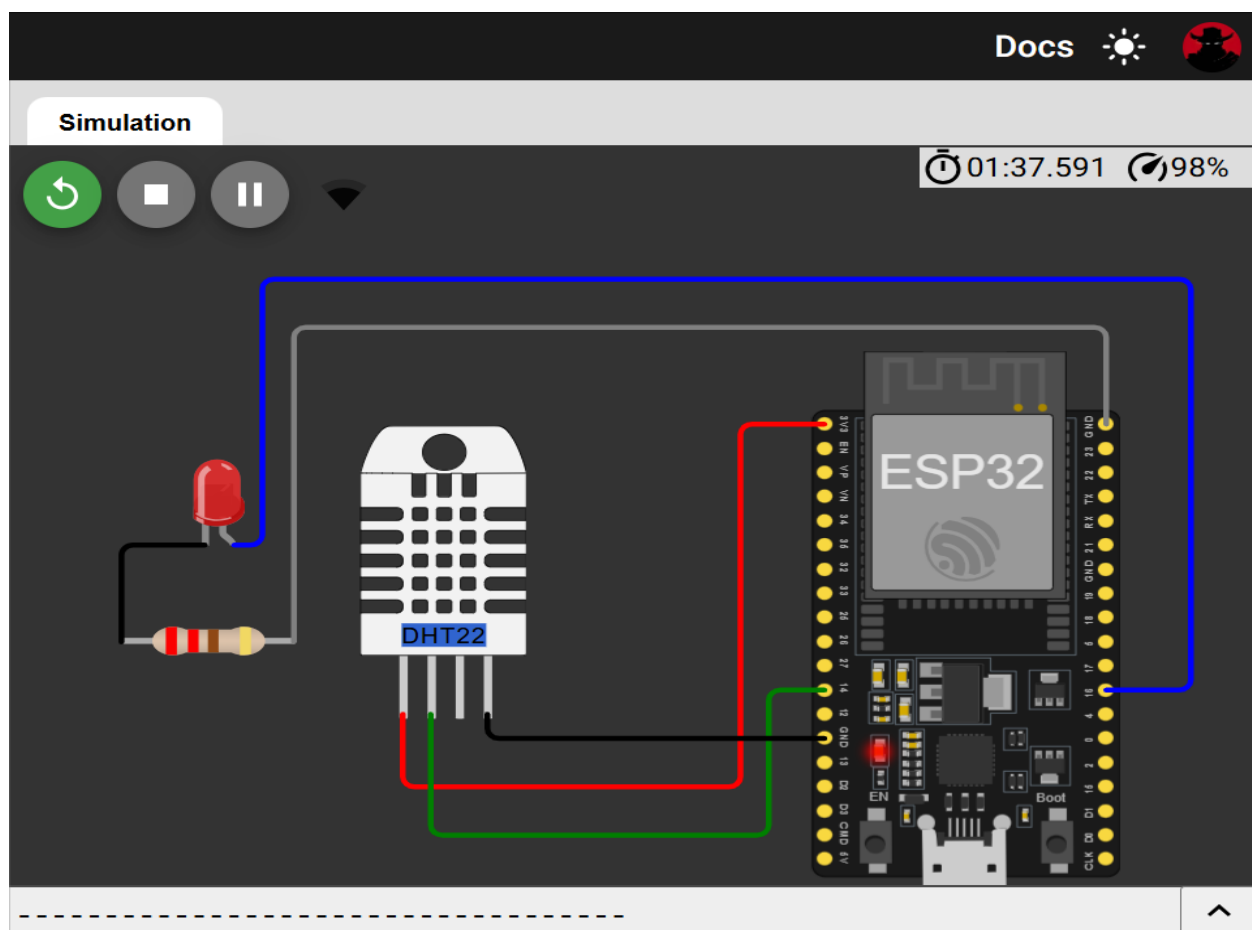
```



```

76 // Send data to ThingSpeak
77 ThingSpeak.setField(1, temperature);
78 ThingSpeak.setField(2, humidity);
79
80 int response = ThingSpeak.writeFields(channelID, writeAPIKey);
81
82 if (response == 200) {
83     Serial.println("📡 Data successfully sent to ThingSpeak");
84 } else {
85     Serial.print("❌ Error sending data. HTTP code: ");
86     Serial.println(response);
87 }
88
89 Serial.println("-----");
90
91 // ThingSpeak requires 15 seconds delay
92 delay(15000);
93 }

```



Connecting to WiFi.....

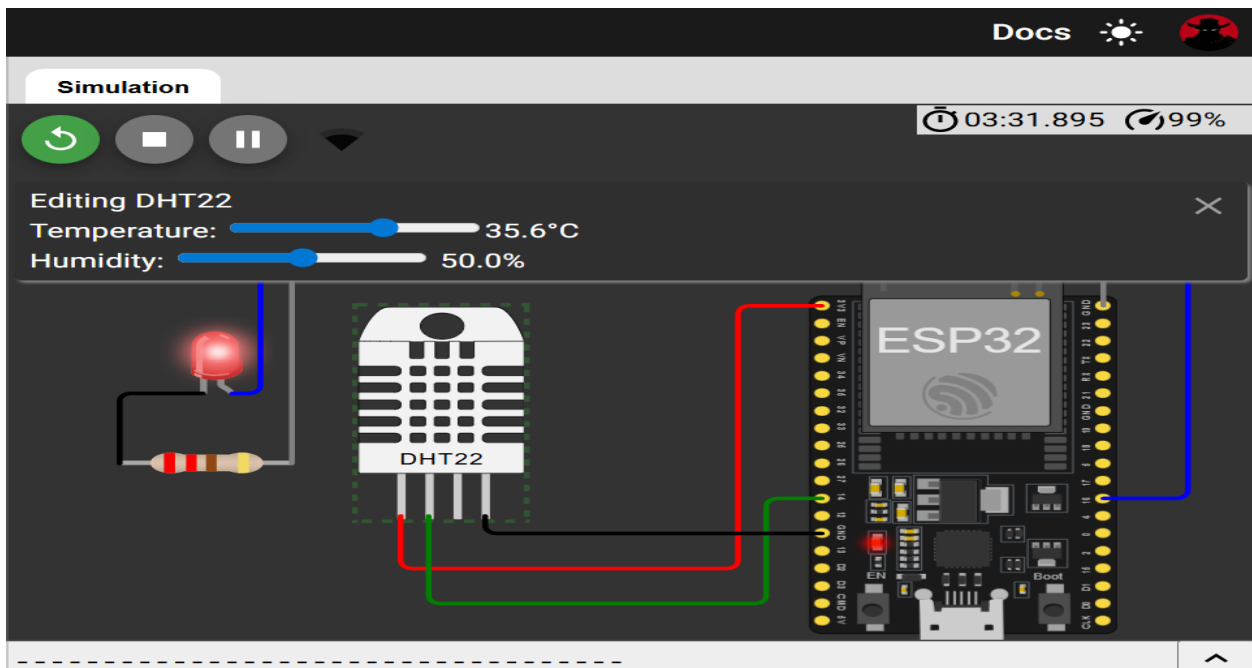
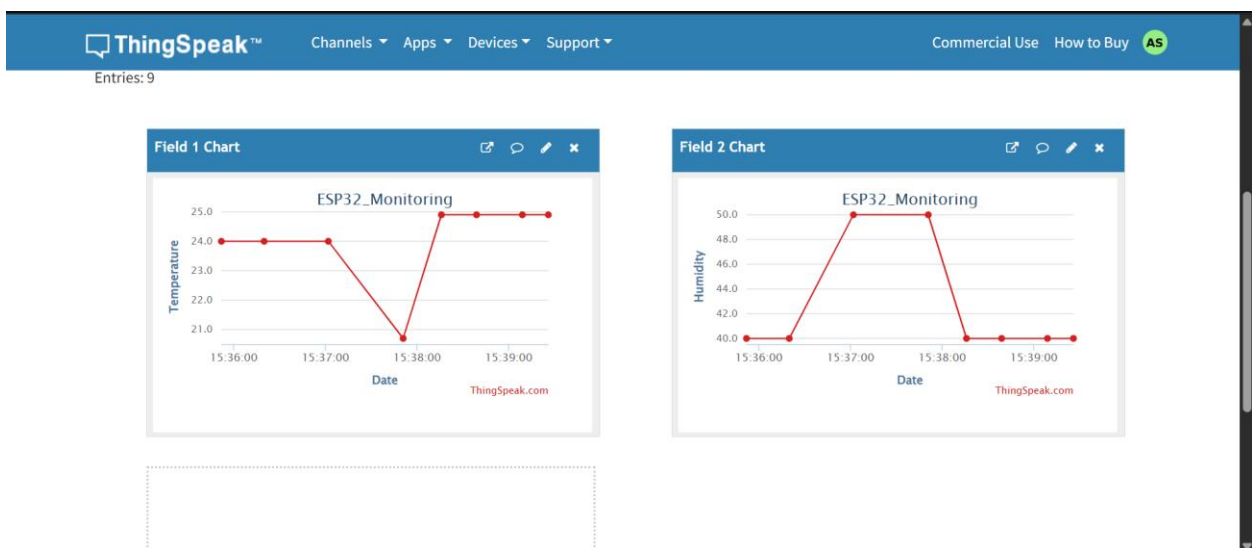
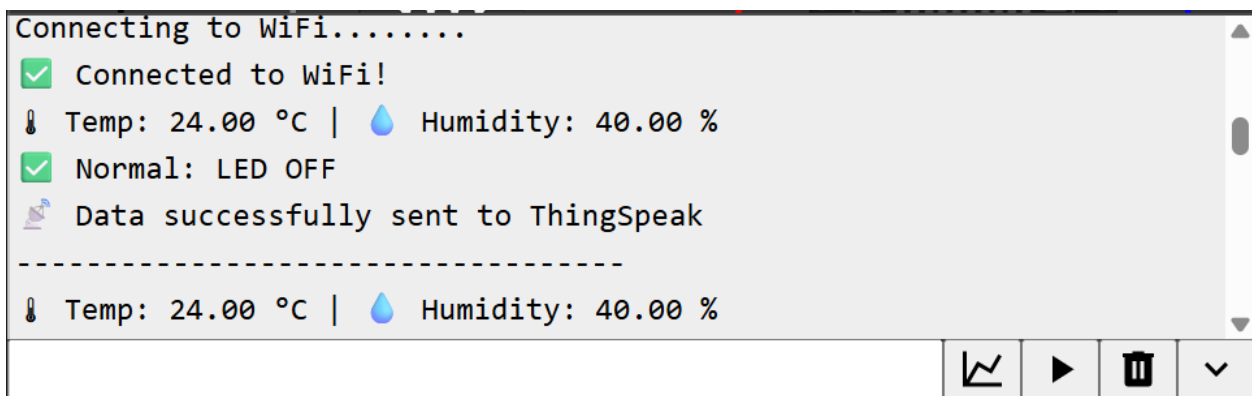
✓ Connected to WiFi!

🌡 Temp: 24.00 °C | 💧 Humidity: 40.00 %

✓ Normal: LED OFF

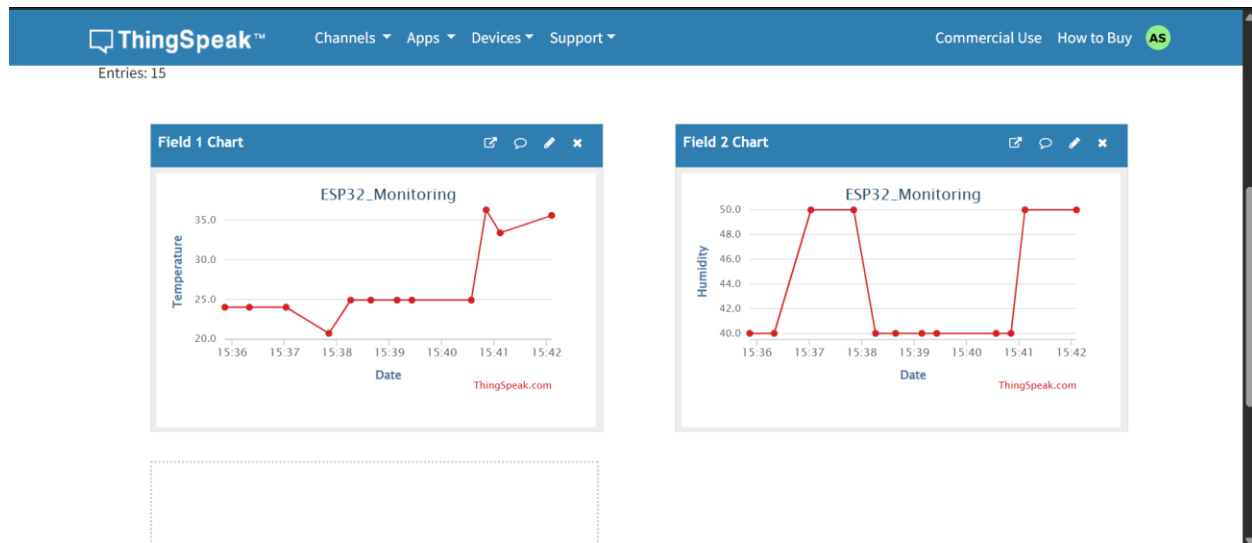
📶 Data successfully sent to ThingSpeak

🌡 Temp: 24.00 °C | 💧 Humidity: 40.00 %







Student Name		LAB Rubrics	CLO3, P5, PLO5
		Total Marks	10
Registration No		Obtained Marks	
		Teacher Name	Dr. Syed M Hamedoon
Date		Signature	